

General Features

The i205 is an intelligent relay output module for the ionia™ system. i205 has 8 relay outputs separated into 4 SPST and 4 SPDT relay. One SPST and one SPDT can be chosen to be bistable prior to ordering .

The configuration registers and all input/output states are accessible by the central unit through a fast and secure point-to-point connection (duagon serial star slave)

i205 is directly supplied by the backplane with no need for an additional DC/DC converter.

A built-in supervisory mechanism called «Condition Monitoring» can detect, record and report any off-limit usage, such as overtemperature, relay defect, etc. as well as selected user-defined values.

Configuring the i205 is done easily and time saving by using one of the five methods defined by IEC 61131.

The i205 is designed for harsh rolling stock environments and is fully compliant to EN 50155.

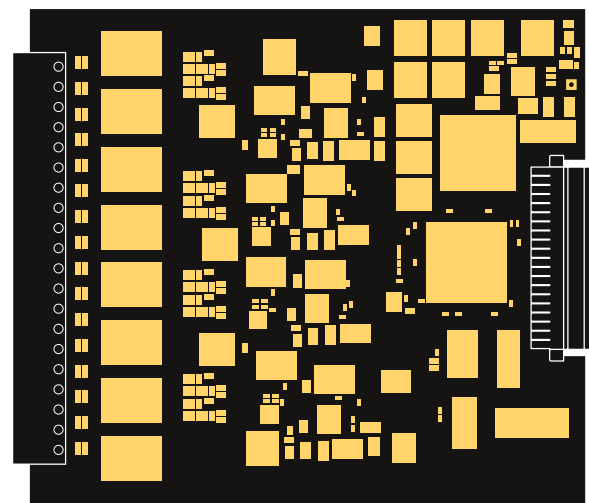
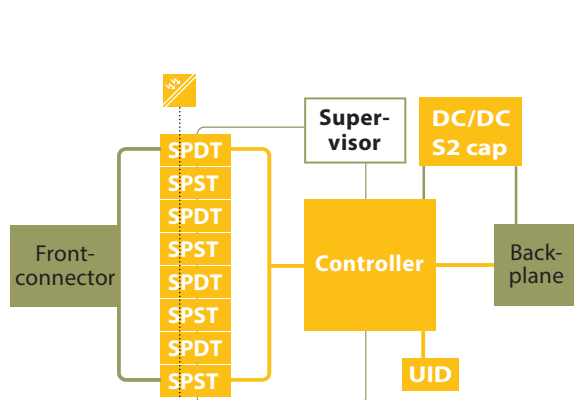
Key Benefits

- Switching free floating potential of any voltage
- Mixed SPST and SPDT relay gives full flexibility
- Optionally bistable relays available
- Defined status in case of an unpowered module
- Configurable status LEDs on each channel for easy diagnostics
- No capacitor replacement necessary
- No need for physical addressing (duagon serial star controller)
- Complies to EN 50155, EN 50121, IEC 61373
- Unique identification number for automatic replacement recognition

Application Examples

- Switching potential free signals
- Eight switches, each switching different Voltages

i205 Hardware Architecture



Life Cycle Cost

The i205 has been strictly designed without electrolytic capacitor to avoid service cost for changing capacitors. The use of an FPGA with included duagon soft-IP minimizes

the risk and costs in case of component obsolescence. Any firmware upgrade can be done via the central unit, without the need of removing the device.

Technical Data

Output Channels	– 4 SPST monostable relay outputs – 4 SPDT monostable relay outputs – Defined fail safe operation on every output channel – Switching Power: 4A @ 250Vac, 0.5A @ 110Vdc, 4A @ 24Vdc – 1500 Vac galvanic isolation to internal logic – 1000 Vac dielectric strength between open contacts – Mechanical endurance: min. 1'000'000 cycles	Diagnostic/Service	– Configurable diagnostic LEDs for each channel indicating the status of the channel or a user-defined functionality – LEDs for visual status information (power on, running, communicating, failure, I/O, ...) – Diagnostic information readable through the central unit's service interfaces <i>(see i101/i102/i103 for further information about these possibilities)</i>
Output Channels Option	– 3 / 1 SPST bistable/monostable relay output – 3 / 1 SPDT bistable/monostable relay output	Power Consumption	– $P_{max} < 2\text{ W}$ (internal logic only + relay coil) – Supports sleep mode
Supply Voltage	– Fully supplied by the backplane – Interruption class S2 internally with ceramic capacitors	Operating Conditions	– Ambient temperature: –40 to +85 °C (EN 50155, Class TX) – Relative humidity: 75%, max 95% for 30 days per year (conformal coating) EN 60068 – Shock and vibration: According to IEC 61373 category 1, class B – EMI: According to EN 50121 and EN 50155
Programming	– Programmable by the central unit using IEC 61131 LD, FBD, ST, IL and SFC – Firmware update handled by the central unit	Dimensions/Weight	– 3TE × 92 × 86 mm, fitting the ionia™ housing – 120 g
Security / Watchdog	– External supervisor temperature monitoring → «Condition Monitoring» – Communication Link with watchdog supervision	Environment	– Fully compliant to RoHS and REACH
Backplane Communication	– 100 Mbit duagon serial star slave – Secure communication protocol		

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Order Code and HW Options

Product Type: **i205** -

EXAMPLE:
i205-M

Relay type

8 monostable	M
6 / 2 mono-/bistable	B

Related Documents

Data Sheet i205  d-006883-nnnnnn.pdf
available at www.duagon.com

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